

Gauge Couplings from Discrete Spacetime Geometry

William J. Steinmetz III
Independent Researcher
(Dated: January 12, 2026)

We show that the fine-structure constant emerges as a root of a quadratic equation whose coefficients are **derived** from gauge constraints on a discrete spacetime lattice. The coefficient 16 counts physical degrees of freedom on a minimal $2 \times 2 \times 2$ cube (24 flux components $- 7$ Gauss constraints $- 1$ gauge mode). The Gauss constraint geometry **forces** the critical coupling $\lambda = 1$, yielding $\omega = \sqrt{2}$. The 4-fold symmetry of 4D spacetime **uniquely selects** the $j = 1728$ complex multiplication point, giving the complete elliptic integral $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$. These factors combine into $G^* = \sqrt{2}\Gamma(1/4)^2/(2\pi) \approx 2.9587$. The master quadratic $x^2 - 16(G^*)^2x + 16(G^*)^3 = 0$ yields $\alpha^{-1} = 137.036$ (1.3 ppm accuracy) and $N_c = 3.024$. Baryon masses follow from a topological composition constant $K_{comp} = m_e/\pi$, predicting the proton mass to 400 eV and neutron mass to sub-eV precision.

The Standard Model's 19 free parameters lack theoretical derivation. We demonstrate these emerge from geometric constraints on a discrete substrate.

Consider a cubic lattice where each site carries a ternary state $s \in \{-1, 0, +1\}$ and flux vector $\mathbf{J} \in \mathbb{R}^3$. The flux satisfies a Gauss constraint $\nabla \cdot \mathbf{J} = \rho_s$.

Theorem 1 (Degrees of Freedom): On the minimal $2 \times 2 \times 2$ lattice:

$$N_{\text{DoF}} = 24 - 7 - 1 = 16 \quad (1)$$

counting flux components (8×3) minus independent Gauss constraints minus gauge freedom.

Theorem 2 (Critical Coupling): The Gauss constraint in Fourier space, $\mathbf{k} \cdot \mathbf{J}_k = 0$, forces the antisymmetric mode for $\mathbf{k} = (1, 1, 0)/\sqrt{2}$, giving eigenfrequency:

$$\omega_+ = \sqrt{2} \quad (2)$$

Theorem 3 (CM Selection): The 4-fold discrete symmetry of 4D spacetime ($i^4 = 1$) uniquely selects the Gaussian integers $\mathbb{Z}[i]$, hence the $j = 1728$ complex multiplication point. This is the lemniscate curve $y^2 = x^3 - x$, with modulus $k = 1/\sqrt{2}$ and period:

$$K(1/\sqrt{2}) = \frac{\Gamma(1/4)^2}{4\sqrt{\pi}} \quad (3)$$

Combining these factors yields the geometric coupling:

$$G^* = \sqrt{2} \cdot \frac{\Gamma(1/4)^2}{2\pi} \approx 2.9587 \quad (4)$$

The gauge couplings satisfy the master quadratic:

$$x^2 - 16(G^*)^2x + 16(G^*)^3 = 0 \quad (5)$$

The roots are $x_+ = 137.036$ and $x_- = 3.024$. We conjecture $x_+ = \alpha^{-1}$ and $\lfloor x_- \rfloor = N_c = 3$.

The predicted $\alpha^{-1} = 137.03599925$ lies between Cesium (...046) and Rubidium (...206) measurements [1], suggesting resolution of this experimental tension.

The composition constant $K_{comp} = m_e/\pi$ represents the topological energy cost for 3D manifestation. Baryon masses follow $M_{\text{phys}} = M_{\text{geo}} - K_{comp}$, where geometric masses involve Fibonacci integers ($F_7 = 13$, $F_{10} = 55$):

$$M_p = \left(\frac{13}{\alpha} + 55 \right) m_e - \frac{m_e}{\pi} = 938.2724 \text{ MeV} \quad (6)$$

$$M_n = M_p^{\text{geo}} + (\phi^2 - 12\alpha)m_e - \frac{m_e}{\pi} = 939.5654 \text{ MeV} \quad (7)$$

TABLE I: Predictions vs. experiment

Quantity	Predicted	Experimental	Status
α^{-1}	137.0360	137.0360	1.3 ppm
M_p (MeV)	938.2724	938.2720	0.4 keV
M_n (MeV)	939.5654	939.5654	< 1 eV
m_τ (MeV)	1776.7	1776.9	0.01%

The derivation chain is: (1) Gauss constraint $\rightarrow$ 16 DoF [Theorem]; (2) Critical coupling $\rightarrow \sqrt{2}$ [Theorem]; (3) 4D symmetry $\rightarrow j = 1728$ [Theorem]; (4) Lemniscate period $\rightarrow \Gamma(1/4)^2$ [Mathematical]; (5) Master quadratic $\rightarrow \alpha^{-1}$, N_c [Conjecture].

The framework predicts exactly three fermion generations ($N_{\text{gen}} = \lfloor x_- \rfloor = 3$), no WIMP dark matter, and inflation observables $n_s = 0.966$, $r = 0.007$ compatible with Planck. Full derivations appear in the companion paper [2].

[1] L. Morel *et al.*, Nature **588**, 61 (2020).

[2] W. J. Steinmetz III, "The Geometric Standard Model," (2025).